PHY304 – Lab 5

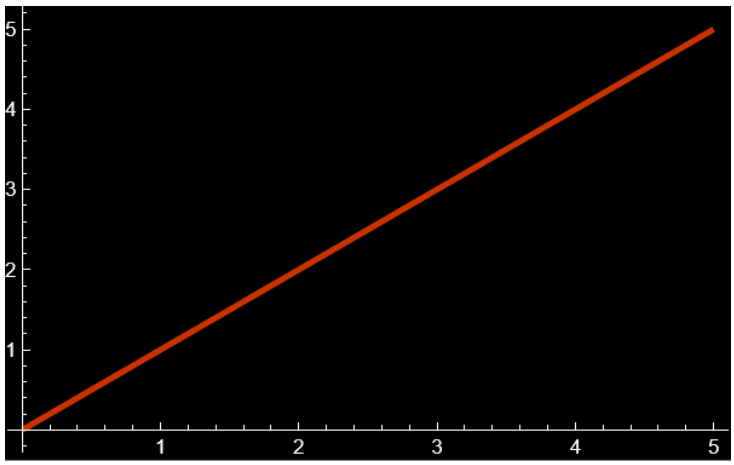
Code and outputs

Cell 1

Input

```
Plot[y=x,{x,0,5}]
```

Out[18]=



Cell 2

Input

```
Clear[f, α, a, ω]
f[t] := α Log[a t]*Cos[ω t]*Sin[ω t]
D[f[t], t]
D[f[t],{t,2}]
```

Out[21]=

$$-\alpha\,\omega\,\log(a\,t)\,\sin^2(t\,\omega)+\alpha\,\omega\,\log(a\,t)\,\cos^2(t\,\omega)+\frac{\alpha\,\sin(t\,\omega)\,\cos(t\,\omega)}{t}$$

Out[22]=

$$\alpha\,\cos(t\,\omega)\left(\omega^2\,(-\log(a\,t))\,\sin(t\,\omega)-\frac{\sin(t\,\omega)}{t^2}+\frac{2\,\omega\,\cos(t\,\omega)}{t}\right)-\alpha\,\omega^2\,\log(a\,t)\,\sin(t\,\omega)\,\cos(t\,\omega)$$
$$-2\,\alpha\,\omega\,\sin(t\,\omega)\left(\omega\,\log(a\,t)\,\cos(t\,\omega)+\frac{\sin(t\,\omega)}{t}\right)$$

Cell 3

Input

```
Clear[f, g, h, w]
w[x_] := f[x]g[x]h[x]
D[w[x], x]
```

Out[25]=

$$g(x)\ h(x)\ f'(x) + f(x)\ h(x)\ g'(x) + f(x)\ g(x)\ h'(x)$$

Cell 4

Input

```
ClearAll[g, l, ω, sol]
g= 9.8
l= 10
ω:= Sqrt[g/l]
sol = NDSolve[{phi''[t] == -ω^2*Sin[phi[t]], phi[0] == 0, phi'[0] == Pi/6}, phi, {t, 0, 10}]
Plot[Evaluate[phi[t] /. First[sol]], {t, 0, 10}]
```

Out[27]=

9.8

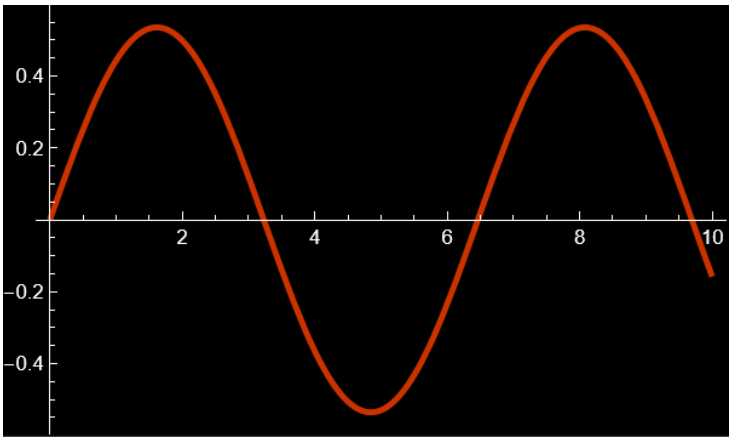
Out[28]=

10

Out[30]=

{{phi → InterpolatingFunction []}}

Plot



Cell 5

Input

```
Graphics3D[{Grey,Cylinder[{{0,0,0},{0,0,2}},(0.05)],Yellow, Cylinder[{{0,-0.1,0},{0,0.1,0}},(1/2)]}
```

Graphics3D

